Trenitalia
ETR1000 VZI-6

Battery

Revision	Author	Date	Revision Description
00.1	G. De Marco	28/08/2023	Draft Release Platform-.Zefiro-Europe/2F_04.02.-.Battery-VF.-.Zefiro-Europe/Battery System-VF.CONCEPT, baseline 2.6, doc 3EST000225-7908_B reviewed for VZI-6
00	G. De Marco	18/10/2023	Issue for VZI-6 PDR
1.01	G. De Marco	22/10/2024	Mod. req 2F_04.02.Zefiro-Europe.CONCEPT.319 (minimum voltage from 77V to 85V)
01	G. De Marco	08/01/2025	Issue for functional BL3
2.01	G. Spina	30/05/2025	Updated document with multiple composition requirements: - New Requirement: HR01 - Modified Requirement CONCEPT.187, TRS.384
02	G. De Marco	22/07/2025	Issue for functional BL4

Content

1 Introduction 4

1.1 Train Configuration 4

1.2 Definitions and abbreviations 4

2 System Overview 5

2.1 System Configuration 5

3 Functional requirements 6

3.1 Failure cases battery system 8
3.1.1 Normal condition 8
3.1.2 Failure cases 8
3.1.3 Failure detection 8
3.1.4 Failure handling 8

3.2 Function Battery Isolation 9
3.2.1 Function Design 9
3.2.2 Requirements 9

3.3 Battery Contactors 11
3.3.1 Failure modes 11
3.3.2 Activation 11
3.3.3 Deactivation 12
3.3.4 Battery charging contactor 14

3.4 Load Shed Contactors 16
3.4.1 Failure modes 16
3.4.2 Activation 16
3.4.3 Deactivation 17

3.5 DC Busline Distribution 19
3.5.1 Connection of loads to battery bus 19

3.6 Battery Charging 21
3.6.1 Failure modes 21
3.6.2 Start and Operation of Battery Charger 22
3.6.3 Dead Battery Start 23
3.6.4 Measuring of battery temperature by battery charger 24

4 Signal Interface 25

4.1 I/O Interface 25
4.1.1 Supplier system 25
4.1.2 TCMS SW 25

4.2 Bus Interface 26

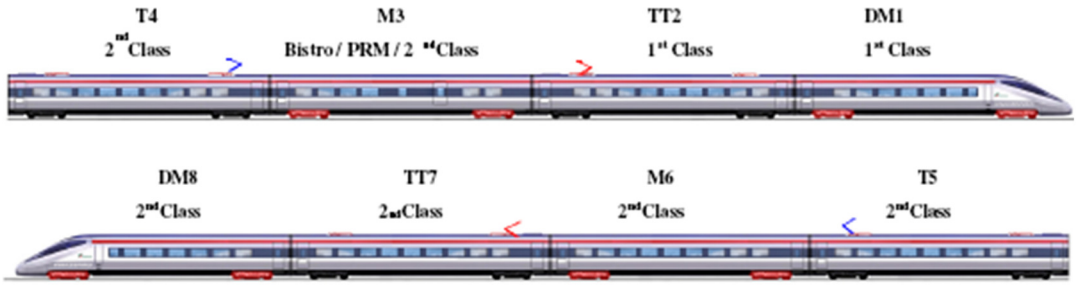
1 Introduction

This document is a function specification for the control of the *110VDC battery* system for V300ZEFIRO for customer Trenitalia.
The batteries provide functions like:

- Energy storage for starting of onboard systems at begin of operation.
- Low inductive buffer for short term high loads like inrush current from electric DC-drives.
- Emergency supply of DC-consumers in case of missing line voltage.
- Energy system for priority loads.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- •DM = End driving motor car with driver's cab
- •TT = Intermediate trailer car with transformer
- •M = Intermediate driving motor car
- •T = Intermediate trailer car

1.2 Definitions and abbreviations

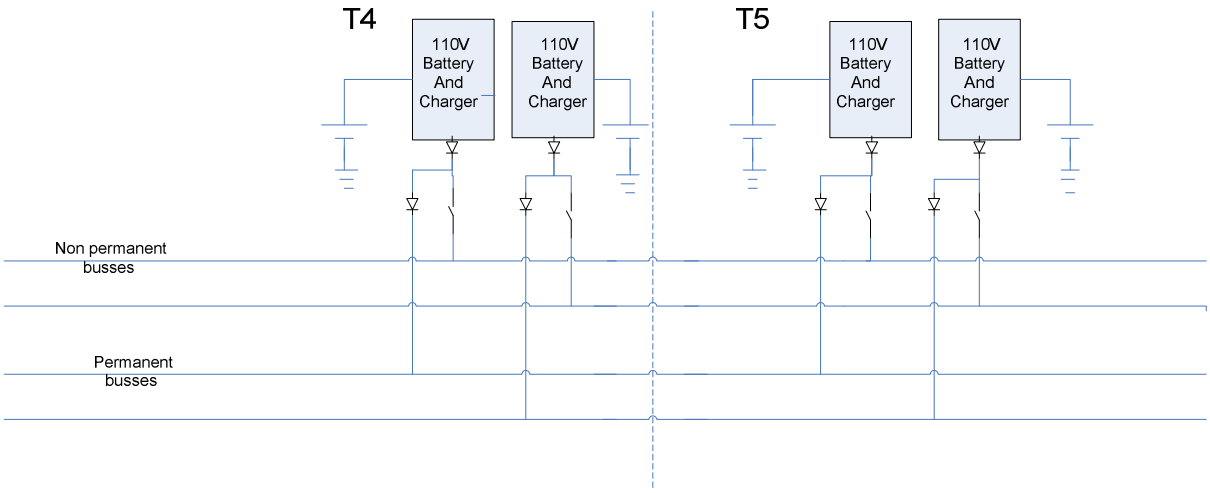
BC1	Battery contactor
BC2	Load shed contactor for high priority equipment
BC3	Load shed contactor for medium priority equipment
CD	Circuit Diagram
DI	Digital Input (to a computer unit)
DD	Drivers desk
DO	Digital Output (from a computer unit)
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
HW	Hardwire
BCM	Battery charging management
SUP	Supplier
MCB	Mini Circuit Breaker
MCTC	Manual Conventional Train Control
MVB	Multifunctional Vehicle Bus
OTC	Ordinary train control
PB	Push button
TBD	To Be Defined

2 System Overview

2.1 System Configuration

General:

The nominal voltage of the battery system is 110VDC.
There are four batteries and four battery chargers mounted in the train, to two battery buses that go through the whole train. Approximately half of the load is connected to each bus.
There are two battery banks located in each T4, and T5 car.
The battery chargers are capable of keeping the batteries fully charged and to supply the different loads on the battery buses with necessary power.
Each battery charger converts three phase power to 110VDC.
The batteries and the battery chargers will be connected in parallel and shall be used to supply the battery buses, start up/priority buses through the train.
The contactors and miniature circuit breakers are normally single poled in this DC-system.
The system is direct grounded (TT-system) and connected to ground in T4-car.
The fundamental task of the battery system will be to generate and distribute a stable 110VDC uninterrupted power supply voltage to all the battery loads in the train and to charge the batteries.
It is possible to disconnect/isolate manually both poles on each battery completely.
Each battery pole is fused.

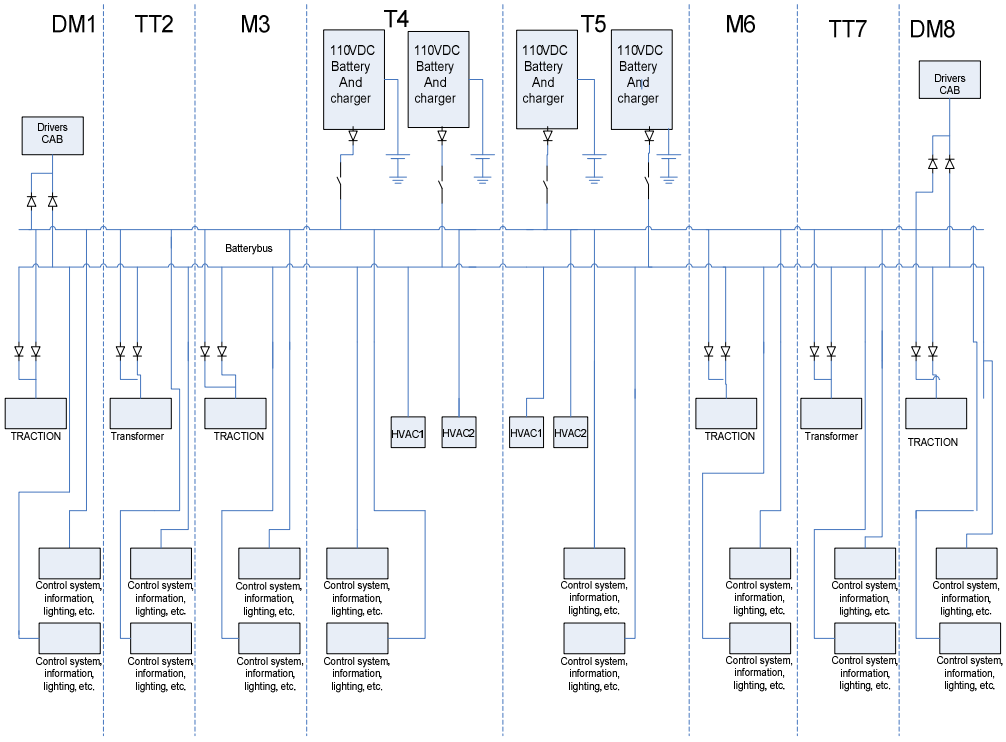


3 Functional requirements

Operational functions:

The following functions are to be included in the battery system:

- Monitoring of the battery voltage
- Monitoring of the battery disconnectors
- Control of battery contactors
- Monitoring of battery contactors and motor protections
- Monitoring of miniature circuit breakers in the battery system. (Normally supervised cabinet for cabinet).
- Monitoring of DC/DC converters in the drivers desk
- Control and monitoring of the battery chargers
- The batteries are equipped with fuses
- Each battery can be disconnected from the battery and permanent bus by a two-pole disconnector with interlocks /supervision to TCMS.
- The batteries and battery chargers are connected to the two battery buses. In addition to the diodes the permanent loads are protected by single pole miniature circuit-breakers.
- It is not possible to start the train unit if the batteries are disconnected and there is no external supply.



(Figure is an example of how different consumers can be supplied)

Normal operating scenarios

The normal operating scenario for the battery system, start up when the train is activated as follows:

Each batterybus connects to loads by the loadshed contactors. (Two contactors for 60min disconnection at no charging, one for each batterybus. Two contactors for 30min disconnection at no charging of the batteries, one for each batterybus. disconnection at no charging of the batteries, one contactor for each bus.)

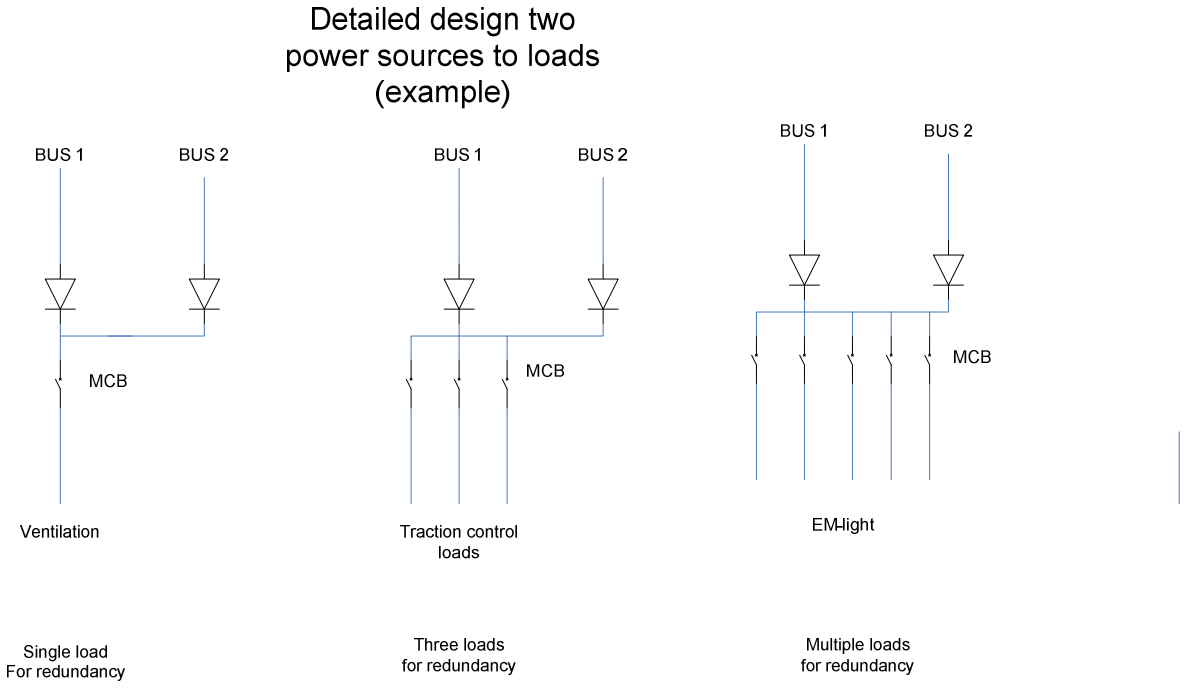
- Battery supply (BC1) contactor activates.
- Battery charger starts when the three-phase voltage is present on the respective car.
- Battery load (BC2) contactor activates
- Battery load (BC3) contactor activates
- The battery charger is capable of keeping the battery fully charged and to supply the loads with necessary power.

Abnormal operating scenarios

The condition of the battery system voltage is monitored at start up and during operating**.** If the discharge continues too long, the monitoring functions will stop the discharge by opening contactors in order to preserve enough energy for the start of the train systems. The battery temperature is measured by the battery charger in order to adapt the charging conditions according to the actual battery temperature**.** The batterybuses is TT, direct connected to ground in the T4 car, T5 has also equipment for grounding, but it is not connected.

It shall be possible to automatically disconnect battery loads with lower priority using the loadshed contactors. The loadshed contactors shall be closed when any battery charger is ON.

Example: different connections to dual DC sources.



3.1 Failure cases battery system

3.1.1 Normal condition

The train has 2 battery-buses with 2 batteries and battery chargers connected to each bus. The loads are supplied from both buses via diodes and with this design the loads are supplied even if one bus fails.

3.1.2 Failure cases

1. One battery disconnected via the isolation switch
2. Two batteries disconnected via the isolation switch
3. One battery fuse melted
4. Two battery fuses melted
5. One battery contactor not closed
6. Two battery contactor not closed
7. One battery charger not working
8. Two battery charger not working
9. Load shed contactors failed
10. One DC bus shut down
11. If 3 battery chargers or more has failed

3.1.3 Failure detection

1,2. Position of the disconnectors monitored. An event shall be set when open
3,4. Fuses are monitored when fuse is blown an event shall be set.
5,6. When battery contactor status and request from pushbutton to activate not align, an event shall be set.
7,8. When battery charger is not active an event shall be set.
9. When load shed contactors status and request from pushbutton to activate not align, an event shall be set
10. Short circuit on the bus or position 2,4,6,8 from above.
11. An event shall be set when battery charger fails

3.1.4 Failure handling

1,3,5,7. No impact on train level, redundancy if one part fails. Repaired in two days.
2,4,6,8. Loads needs to be reduced. Repaired in 2 days.
9. No impact on train level, all loads are still supplied. Repaired in 2 days
10. Loads needs to be reduced. Repaired in 2 days.
11. 3 battery chargers or more has failed the train shall be withdrawn from service.

3.2 Function Battery Isolation

3.2.1 Function Design

It is possible to manually isolate both poles on each battery pack by disconnecting it from all the loads including the battery charger. Each battery pole is protected with a thermal fuse.
All battery disconnectors in the train have to be opened manually before maintenance work on the battery or the battery circuits, or before the train are taken out of service for a long period.
The train is equipped with 4 workshop sockets which aim to supply the train when the batteries are not available or for charging of each battery whenever the train is in the depot.

For each battery disconnector a fault event is given to indicate if it is open. As the battery contactors (BC1) are feeding the TCMS system no fault indications or supervision is available when all BC1 are open as the TCMS is shut down

3.2.1.1 Normal condition

During normal operation the battery disconnectors are closed and will be opened only during maintenance, long time parking or at fault. If short circuit occurs on the DC bus the battery thermal fuse melts.
When the train is activated, an event on IDU shall be set if the battery disconnector is not closed.
(Note: All battery disconnector can not be switched off, the train can not be activated when TCMS is not switched on)

3.2.1.2 Failure Cases

1. Auxiliary contact on battery disconnector does not close, when the battery disconnector is closed
2. Auxiliary contact on battery disconnector is closed, when the battery disconnector is open.

3.2.1.3 Failure detection

1. An event on IDU shall be set when the battery disconnector is not closed. The auxiliary contact status shall be compared with the battery voltage sent on IP-bus from battery charger.
2. When auxiliary contact and disconnector are not align. The auxiliary contact status shall be compared with the battery voltage sent on IP-bus from battery charger. If status for battery voltage from battery charger and battery disconnector are not the same, an event shall be set.

3.2.1.4 Failure handling

1. Repaired in 2 days.
2. Repaired in 2 days.

3.2.2 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.364
Description	It shall be possible to disconnect/isolate manually both poles on each battery completely from the battery bus and from the battery charger.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.72	The battery shall be connected to the battery charger and the battery bus via a manual isolation switch.	Derived Requirement	4S_04.04.03.-.Battery Boxes		N/A
2F_04.02.Zefiro-Europe.CONCEPT.520	The manual isolation switch has 3 positions: 1) normal service: battery connected, train workshop socket isolated; 2) workshop battery charging: battery connected, train workshop socket connected; 3) maintenance mode: battery isolated, train workshop socket connected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.521	Each battery can be supplied from its own train workshp socket, placed close to manual isolation switch.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_04.02.Zefiro-Europe.TRS.366
Description	The position of the battery isolator switch shall be supervised
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.252	If the battery isolation switch is not in duty position, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.254	The status of the battery isolation switches (open or closed) shall be shown on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.493	The status of the battery isolation switch "Battery isolation switch open" shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_04.02.Zefiro-Europe.TRS.650
Description	Each battery pole shall be fused.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.200	The battery fuses on each battery pole shall be placed as close to the battery cells as possible	Derived Requirement	4S_04.04.03.-.Battery Boxes		N/A
2F_04.02.Zefiro-Europe.CONCEPT.263	The battery fuses shall be designed to allow supervision of the fuses.	Derived Requirement	4S_04.04.03.-.Battery Boxes		N/A
2F_04.02.Zefiro-Europe.CONCEPT.477	The status "NO battery fuse tripped" shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.264	If a battery fuse is tripped an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.3 Battery Contactors

3.3.1 Failure modes

3.3.1.1 Normal condition

All loads in the train, except the start up/permanent buses shall be direct connected to one of the battery buses. Each battery and the battery charger is connected to the battery bus with the activation pushbutton of the single pole battery contactors.
Normal condition all battery contactors are closed.

3.3.1.2 Failure detection

1. When battery contactor status and request from pushbutton to activate not align, an event shall be set.
2. See also chapter 3.1

3.3.1.3 Failure cases

1. If battery contactors does not open

3.3.1.4 Failure handling

1. If battery contactor does not open, the battery disconnector has to be opened manually for the actual battery. Repaired in 2 days

3.3.2 Activation

3.3.2.1 Functional design

The primary battery contactor is controlled by relay logic, and this logic requires two separate signals, one to close this contactor and one to open it. The battery contactor is opened when the "open battery contactor" from TCMS DO output is set, or when the battery disconnector opens.

The pushbutton in DM car electrical cabinet will activate the train by closing the battery contactors in the train.

3.3.2.2 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.378
Description	The contactor connecting battery (and charger) to the battery bus shall be closed on a manual order.
Safety level	SIL 0

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.02.Zefiro-Europe.CONCEPT.266	All primary battery contactors (including multiple units) shall close when the "Close Battery Contactor" button is pushed	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

ID	2F_04.02.Zefiro-Europe.TRS.390
Description	The position of all battery contactors including the contactors for load shedding shall be supervised
Safety level	SIL 0

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.02.Zefiro-Europe.CONCEPT.322	If BC1 does not close on order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.321	If BC1 does not open on order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_04.02.Zefiro-Europe.CONCEPT.492	The status "BC1 closed" shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_04.02.Zefiro-Europe.TRS.368
Description	The actual status of each battery circuit breaker (open / closed) shall be available on the IDU
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.187	The IDU shall show the status for each BC1 (open/closed) in the trainset	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

3.3.3 Deactivation

3.3.3.1 Function Design

The primary battery contactors is opened by manual order from the driver, via a pushbutton, or by the train control if any low levels of battery voltage or discharge time has exceeded.

3.3.3.2 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.384
Description	The contactor connecting battery (and charger) to the battery bus shall open on manual order (in multiple composition, also issued from remote train) or automatic battery protection order, but only if the train is in a safe condition.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.268	Sequence to open BC1 shall be activated if the "open battery contactor" is pushed, and all of the following conditions are fulfilled: – No cab is active – Speed of train is 0km/h – No pantograph is raised. – No battery charger is charging	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.HR01	In Multiple Unit if the opening sequence of BC1 is activated , TCMS shall command the opening sequence of BC1 on the remote train	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.362	The pushbutton for "open battery contactor" shall have the following behavior: – NOT LIT: Manual opening of battery contactor is not allowed – LIT: Manual opening of battery contactor is allowed – FLASHING: Sequence to open battery contactor is activated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.494	The indication lamp of the Battery contactor push button shall be connected to a TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.276	Since 400V is not longer available the BC1 shall be opened after 180 minutes if the train is at standstill.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.02.Zefiro-Europe.CONCEPT.320	The sequence to open BC1 shall be as follows: 1. An order to the IDU:s, CCU-D:s and other systems (for example Line voltage) to shut down and to prevent live voltage activation shall be sent. 2. After 32 seconds the Load Shed Contactor BC3 shall open. 3. After 34 seconds the Load Shed Contactor BC2 shall open. 4. After 55 seconds an event shall be set 5. After 60 second delay the BC1 shall open	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.345	Once the open BC1 sequence is activated it shall not be possible to reset the sequence.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.498	The open order to battery contactor BC1 shall be connected to a TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_04.02.Zefiro-Europe.TRS.775
Description	If no battery charger is charging and the train has been stationary for a certain time the battery contactors shall open to prevent deep discharge.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.319	The sequency to open BC1 shall be activated if ALL of the following conditions are fulfilled for 10 minutes: – No auxiliary voltage present – Train is at stand still – No cab is active. The sequency to open BC1 shall be also activated if ALL of the following conditions are fulfilled for 1 minute: – voltage of both battery bus falls below 85V – Train is standstill – Train is ON since at least 5 min – No auxiliary voltage present	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.02.Zefiro-Europe.TRS.396
Description	Supervison of low battery voltage shall be used to allow normal shut down before the battery is deep discharged.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.272	If the highest battery voltage (max of the 2 battery bus) drops to a value less than or equal to 86V for more than 3' whithin the first 30' of load shedding a protection sequence shall be started.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.529	When the protection sequence has started an event shall be set with the message: **Battery Low: Ventilation and lights will be switched off within 2 min**.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.530	The protection sequence consists of delivering an untimely (before the 30th minute of loadshedding) BC3 open order. Such protection sequence can be inhibited by the driver by pushing HC-S001 push button. In case the driver interrupts the protection sequence, the BC3 open order will come at the 30th minute as defined with the load shedding.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.531	If the driver inibhits the protection action an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.02.Zefiro-Europe.CONCEPT.273	If the higher battery voltage (max of the 2 battery bus) fall down to 88V for 3' during the last 150' of load shedding, a waning event shall be set to driver. **ATTENTION !!!! The batteries are close to the complete discharge**	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_04.02.Zefiro-Europe.TRS.770
Description	A warning to equipment that needs to be shut down before power off shall be sent prior opening of battery contactor
Safety level	N/A

3.3.4 Battery charging contactor

3.3.4.1 Functional design

Battery charging is managed by an external contactor controlled directly by battery charger logic output. Through the contactor the battery is disconnected from the battery charger once the battery is fully charged.
Battery supplies the DC buses on demand as back-up in case of missing supply from battery chargers or when the battery chargers are overloaded.
The contactor's triggering depends on the battery current measured by the battery charger. The battery current is measured both when battery is in charging and discharging stage.

3.3.4.2 Settings for opening and closing the contactor –K

When the Train DC load exceeds the charger's ability to supply at the time, or in back-up, the bus voltage drops to around its nominal voltage, or zero, causing the forward bias of the diode and then the battery is being discharged.
The regulation of the battery charging contactor is based on a battery current hysteresis. Once the battery discharging current exceeds a certain threshold I2 (=10A +/-2A), the contactor is closed and thus the battery is being charged again until the charging current exceeds I1 (=6A +/-2A) meaning the battery is fully charged. At this stage, contactor is triggered to opening and the battery stays charged till next requesting of discharging.

Nevertheless the contactor shall be opened when the battery temperature is more than 60°C, over temperature protection. Feedback from the contactor shall be available on IP.

The battery contactor is being closed once per day in order to ensure full state of charge of battery.

3.3.4.3 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.787
Description	The battery charger controls the Batt Charg Contactor-K
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.509	Battery charger has a dedicated output to command battery charging contactor and input to read the status	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.510	Batt Charg Contactor-K is placed between battery and battery charger wiring . It is used for disconnect the battery to the battery charger.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.511	Battery charger measures the battery current both in charging and discharging stage	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A
2F_04.02.Zefiro-Europe.CONCEPT.512	When the battery discharging current exceeds I2 (=10A +/-2A) the Batt Charg Contactor-K is closed and then battery is being charged	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A
2F_04.02.Zefiro-Europe.CONCEPT.513	Battery charging contactor is being closed once per day	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A
2F_04.02.Zefiro-Europe.CONCEPT.515	When the battery charging current falls below I1 (=6A +/-2A) and Ubat=Umax the Batt Charg Contactor K is being opened and then battery will	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A

	be discharged if necessary (DC bus overload, backup)				
2F_04.02.Zefiro-Europe.CONCEPT.516	If Batt Charg –K contactor does not close/open on order an event shall be sent	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A
2F_04.02.Zefiro-Europe.CONCEPT.518	Battery charging contactor shall be opened when the battery temperature is more than 60°C	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A

3.4 Load Shed Contactors

The battery system is designed to allow consumers to be disconnected in 3 levels. The primary battery contactor will feed the two separate battery buses. Load shedding contactors (BC2 and BC3) are all connected to the battery bus, but can be controlled to open separately depending on the situation.

3.4.1 Failure modes

3.4.1.1 Normal condition

Load shed contactor are closed for both buses in normal operation

3.4.1.2 Failure cases

1. Load shed contactor does not open when Train control deactivates load shed contactor
2. See also chapter 3.1

3.4.1.3 Failure detection

1. An event shall be set when status and request do not align.

3.4.1.4 Failure handling

1. All load shed contactors are redundant. Repaired in 2 days

3.4.2 Activation

3.4.2.1 Function design

The load shed contactors BC2, BC3 are kept open until battery chargers are running. If a load shed contactor will not close, the loads will be powered from other load shed contactor and no load will be power less due to a single failure.

3.4.2.2 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.320
Description	The distribution of loads shall work according to a hierarchical switch-off management by connecting differnt groups of consumers to different loadshedding groups.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.281	Load shed contactors (BC2 and BC3) on a battery bus shall close when all of the following conditions are fulfilled: any battery charger on or Towing mode switch is selected in : Treno soccorso (C) Treno soccorso (T) 	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.02.Zefiro-Europe.CONCEPT.522	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2F_04.02.Zefiro-Europe.CONCEPT.523	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.02.Zefiro-Europe.CONCEPT.524	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	
2F_04.02.Zefiro-Europe.CONCEPT.495	The close order to Load shed contactors BC2 shall be connected to a TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.496	The close order to Load shed contactors BC3 shall be connected to a TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.497	DELETED	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_04.02.Zefiro-Europe.TRS.390
Description	The position of all battery contactors including the contactors for load shedding shall be supervised
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.301	If BC2 does not close on order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.02.Zefiro-Europe.CONCEPT.305	If BC3 does not close on order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.02.Zefiro-Europe.CONCEPT.306	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.02.Zefiro-Europe.CONCEPT.300	If BC2 does not open on order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.02.Zefiro-Europe.CONCEPT.304	If BC3 does not open on order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.02.Zefiro-Europe.CONCEPT.309	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.02.Zefiro-Europe.CONCEPT.489	The status "BC2 closed" shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.491	The status "BC3 closed" shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.490	DELETED	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.4.3 Deactivation

3.4.3.1 Function design

The load shed contactor is opened depending on discharge time and battery voltage to protect the batteries from deep discharge.

3.4.3.2 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.320
Description	The distribution of loads shall work according to a hierarchical switch-off management by connecting differnt groups of consumers to different loadshedding groups.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.346	BC3 shall open (if its not already open) when BC2 is ordered to open	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.02.Zefiro-Europe.CONCEPT.347	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.02.Zefiro-Europe.CONCEPT.280	Since 400V is not longer available the BC2 shall be opened after 60 minutes.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.02.Zefiro-Europe.CONCEPT.299	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.02.Zefiro-Europe.CONCEPT.279	Since 400V is not longer available the BC3 shall be opened after 30 minutes.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.02.Zefiro-Europe.CONCEPT.302	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.02.Zefiro-Europe.CONCEPT.278	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.5 DC Busline Distribution

3.5.1 Connection of loads to battery bus

3.5.1.1 Function Design

The loads are connected to both battery buses decoupled by diodes.

3.5.1.2 Normal condition

The loads are supplied from both buses, if one DC bus fails the loads still have supply. All loads are protected with Miniature Circuit Breakers (MCBs).

3.5.1.3 Failure cases

1. One DC bus shutdown
2. MCB trip

3.5.1.4 Failure detection

1. If one DC bus is shut down the priority loads are supplied from the other bus, some loads might be reduced. Instant workshop.
2. If mini circuit breaker trips, required redundant loads is cancelled by other circuits.

3.5.1.5 Failure handling

1. See chapter 3.1
2. Repaired in 2 days

3.5.1.6 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.370
Description	Safety-relevant DC consumers shall be connected to a permanent battery bus that is not disconnected from the batteries at shut down .
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.02.Zefiro-Europe.CONCEPT.102	The following safety related consumers shall not be disconnected when the primary battery contactors opens: – Fire safety equipment – Emergency lighting	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_04.02.Zefiro-Europe.TRS.696
Description	All DC mini circuit breakers shall be supervised. A tripped MCB shall be indicated by a diagnostic message indicating the location of the tripped MCB.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.02.Zefiro-Europe.CONCEPT.249	The status switch for all MCBs in a cabinet shall be daisy chained to allow supervision of the MCB status on cabinet level.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.323	If an MCB is tripped an event shall be set indicating in which cabinet the MCB has tripped.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_04.02.Zefiro-Europe.TRS.777
Description	It shall be possible to connect loads in a way that a load is still powered if a failure occur in one of the two separate battery buses.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.486	Loads shall be connected to the contactors according to power balance document sheet battery regime BC3=30min (B/C) BC2=60min (D) BC1=180min (D/E) Permanent loads (E)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.487	When at least two faults that prevents a battery charger to supply the battery bus are present, i.e combination of – BC1 not possible to close – Battery charger not operating with 400V available an event shall be set and the following loads shall be reduced: – The LCD screens PIS system shall be order to standby mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.501	When at least 3 faults that prevents a battery charger to supply the battery bus are present, i.e combination of – Battery charger not operating with 400V available – BC1 not possible to close an event shall be set that the train shall be withdrawn from service.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.02.Zefiro-Europe.CONCEPT.287	The cab HVAC shall be able to operate even if one battery bus is out of order.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.284	All external doors shall be able to operate even if one battery bus is out of order.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.288	The train radio shall be able to operate even if one battery bus is out of order.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.289	All toilets shall be able to operate even if one battery bus is out of order	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.286	The PA part of the PIS shall be able to operate if one battery bus is out of order	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.290	The fire detection and extinguish system shall be able to operate if one battery bus is out of order.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.285	Minimum 50% of internal lightning shall be functional if one battery bus fails	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.283	The HVAC shall be able operate normally even if one battery bus is out of order.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.6 Battery Charging

3.6.1 Failure modes

3.6.1.1 Function Design

The battery charger is providing measured data to the train control system to allow presentation of the status of the dc-supply system in the drivers cab.

3.6.1.2 Normal condition

The consequences can be that battery charger decrease charging according to failed battery temperature. If single failure to one battery bus; this will have no affect on train.

3.6.1.3 Failure cases

1. Battery temperature sensor is failed, and sends a not correct temperature signal to the battery charger.
2. Charging/discharging current missing.
3. Voltage measurement missing.

3.6.1.4 Failure detection

1. The battery temperature shall be available on IDU, and non relevant temperature can be compared according to other battery boxes temperature by driver/service personnel.
2. Current measurement can be compared to the other battery chargers.
3. Voltage measurement can be compared to the other battery chargers.

3.6.1.5 Failure handling

1. Battery temperature sensor needs to be replaced. Repaired in 2 days.
2. Repaired in 2 days.
3. Repaired in 2 days.

3.6.1.6 Battery Monitoring, Battery Management

The battery charging shall be adjusted by the battery charger and be dependent on the battery temperature. The battery charger controls the battery charging by having different outputs to battery and 110V consumers.

3.6.1.6.1 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.348
Description	A battery management system shall monitor – battery voltage – charging and discharging current – battery temperature – battery network voltage These values shall be available on the IDU.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.02.Zefiro-Europe.CONCEPT.142	The Battery charger shall communicate the following values: – Battery voltage – Charging and discharging current – Battery temperature	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A

2F_04.02.Zefiro-Europe.CONCEPT.310	The driver shall be able to monitor the following measured value in the cab: – Battery voltage (for each battery) – Charging and discharging current – Battery temperature – Voltage of the battery bus (for each bus).	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.02.Zefiro-Europe.CONCEPT.478	The driver shall be able to monitor the following measured value in the cab: – Battery voltage (for each battery) – Charging and discharging current – Battery temperature – Voltage of the battery bus (for each bus).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	IDU	N/A
2F_04.02.Zefiro-Europe.CONCEPT.197	If the battery voltage is lower than 20 or higher than 150V the measurement shall be considered faulty and the IDU shall show an indication that the measured value is out of range	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.02.Zefiro-Europe.CONCEPT.348	If the battery voltage is lower than 20 or higher than 150V the measurement shall be considered faulty and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_04.02.Zefiro-Europe.TRS.657
Description	The actual value of the battery voltage shall be available on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.221	The battery voltage for each battery shall be shown on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

ID	2F_04.02.Zefiro-Europe.TRS.656
Description	The charging/discharging current of the battery shall be available on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.224	The battery current shall be shown on the IDU as a negative value at discharge and a positive value at charging.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

ID	2F_04.02.Zefiro-Europe.TRS.658
Description	The battery bus voltage shall be available on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.231	The highest measured voltage of each battery bus shall be shown on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

3.6.2 Start and Operation of Battery Charger

3.6.2.1 Function Design

The battery charger is connected to 3*400V from the auxiliary power bus. As soon as the bus is powered up the battery charger starts charging. Over current, over temperature etc. is handled by the battery charger itself.

3.6.2.2 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.411
Description	Battery charger shall start automatically when input voltage is detected
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.147	When 3*400AC is detected by the battery charger it shall start charging. No enabling signals shall be needed.	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A

ID	2F_04.02.Zefiro-Europe.TRS.639
Description	The battery charger shall at least have following functions: – Charging current and voltage control (acc. charging characteristics) – Battery temperature compensation of charging voltage, for which temperature sensors shall be installed in the battery. – Total output current limitation. – Over temperature protection/indication in charger or battery
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.214	The battery charger shall be equipped with: – Temperature compensation – Current limitation – Over temperature protection	Derived Requirement	4S_04.04.02.-.Battery Chargers		SIL 0
2F_04.02.Zefiro-Europe.CONCEPT.194	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_04.02.Zefiro-Europe.TRS.127
Description	At least the following fault messages shall be provided – battery charger active (able to charge and not faulty)
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.243	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_04.02.Zefiro-Europe.TRS.431
Description	The actual status of the battery charger shall be available on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.234	The IDU shall show one of the following statuses of each battery chargers: – Charging – Not charging – Faulty	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

3.6.3 Dead Battery Start

3.6.3.1 Function Design

The system must be designed to handle a situation where the batteries are completely discharged. If the battery charger is powered the charging of the battery must start regardless of the discharge level of the battery**.**

3.6.3.2 Requirements

ID	2F_04.02.Zefiro-Europe.TRS.415
Description	Start of battery charging with totally discharged battery shall be possible if the train is fed from internal or external three-phase supply
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.157	The battery charger shall be able to charge a totally discharged battery.	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A

3.6.4 Measuring of battery temperature by battery charger

3.6.4.1 Function Design

The battery charger is supposed to handle different battery temperatures and is therefore fitted with temperature inputs that are connected to temperature sensors in the battery itself.

3.6.4.2 Requirement

ID	2F_04.02.Zefiro-Europe.TRS.425
Description	The battery temperature shall be measured and that signal shall be hard-wired from the battery to the battery charger.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.349	The battery shall be equipped with temperature sensors	Derived Requirement	4S_04.04.01.-.Batteries		N/A
2F_04.02.Zefiro-Europe.CONCEPT.479	The battery shall be equipped with temperature sensors	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.02.Zefiro-Europe.CONCEPT.315	The battery temperature shall be measured	Derived Requirement	4S_04.04.02.-.Battery Chargers		N/A

ID	2F_04.02.Zefiro-Europe.TRS.655
Description	The temperature of the batteries shall be available on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.02.Zefiro-Europe.CONCEPT.227	The temperature of each battery shall be shown on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Hardware interface where the sub supplier must provide a hardware connection like a relay output, etc.

Inputs:
Outputs:

4.1.2 TCMS SW

I/O interface where TCMS is connected to the part of the system that is built by separate components and are not part of a sub supplier scope

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_04.02.Zefiro-Europe.CONCEPT.340	Battery bus voltage	A	2F_04.02.Zefiro-Europe.CONCEPT.231 2F_04.02.Zefiro-Europe.CONCEPT.272 2F_04.02.Zefiro-Europe.CONCEPT.273
2F_04.02.Zefiro-Europe.CONCEPT.335	Battery contactor open push button	L	2F_04.02.Zefiro-Europe.CONCEPT.320
2F_04.02.Zefiro-Europe.CONCEPT.336	Battery isolation switch open	L	2F_04.02.Zefiro-Europe.CONCEPT.252 2F_04.02.Zefiro-Europe.CONCEPT.254
2F_04.02.Zefiro-Europe.CONCEPT.337	No MCB tripped	L	2F_04.02.Zefiro-Europe.CONCEPT.323
2F_04.02.Zefiro-Europe.CONCEPT.344	No Battery fuse tripped	L	2F_04.02.Zefiro-Europe.CONCEPT.264
2F_04.02.Zefiro-Europe.CONCEPT.330	BC1 Closed	L	2F_04.02.Zefiro-Europe.CONCEPT.187 2F_04.02.Zefiro-Europe.CONCEPT.321 2F_04.02.Zefiro-Europe.CONCEPT.322
2F_04.02.Zefiro-Europe.CONCEPT.331	BC2 Closed	L	2F_04.02.Zefiro-Europe.CONCEPT.300 2F_04.02.Zefiro-Europe.CONCEPT.301
2F_04.02.Zefiro-Europe.CONCEPT.332	BC3 Closed	L	2F_04.02.Zefiro-Europe.CONCEPT.304 2F_04.02.Zefiro-Europe.CONCEPT.305

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_04.02.Zefiro-Europe.CONCEPT.31	Open BC1	L	2F_04.02.Zefiro-Europe.CONCEPT.320
2F_04.02.Zefiro-Europe.CONCEPT.325	Close BC2	L	2F_04.02.Zefiro-Europe.CONCEPT.280 2F_04.02.Zefiro-Europe.CONCEPT.281 2F_04.02.Zefiro-Europe.CONCEPT.299
2F_04.02.Zefiro-Europe.CONCEPT.326	Close BC3	L	2F_04.02.Zefiro-Europe.CONCEPT.279 2F_04.02.Zefiro-Europe.CONCEPT.302 2F_04.02.Zefiro-Europe.CONCEPT.346
2F_04.02.Zefiro-Europe.CONCEPT.364	Battery contactor push button indication		2F_04.02.Zefiro-Europe.CONCEPT.362

4.2 Bus Interface

Bus interface between the sup supplier equipment and TCMS

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_04.02.Zefiro-Europe.CONCEPT.355	Battery charger is charging	L	2F_04.02.Zefiro-Europe.CONCEPT.243
2F_04.02.Zefiro-Europe.CONCEPT.499	Battery charger is faulty		2F_04.02.Zefiro-Europe.CONCEPT.234
2F_04.02.Zefiro-Europe.CONCEPT.29	Battery voltage	A	2F_04.02.Zefiro-Europe.CONCEPT.221
2F_04.02.Zefiro-Europe.CONCEPT.338	Battery current (+-)	A	2F_04.02.Zefiro-Europe.CONCEPT.224
2F_04.02.Zefiro-Europe.CONCEPT.339	Battery temperature	A	2F_04.02.Zefiro-Europe.CONCEPT.227

TCMS SW to Supplier: